

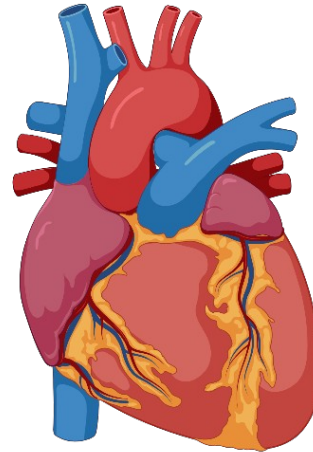
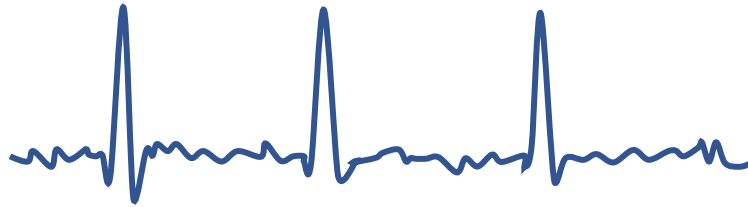
Stroke Prevention

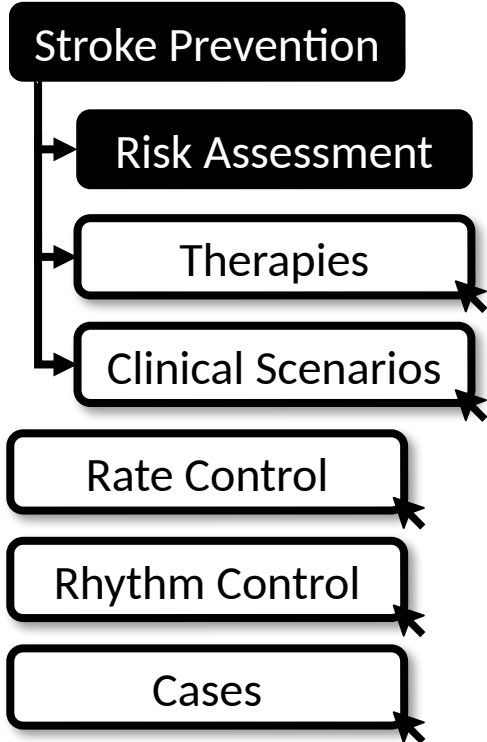
Rate Control

Rhythm Control

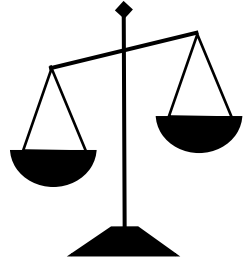
Cases

Chronic Management of Atrial Fibrillation





Stroke Prevention: Risk Assessment



Stroke risk tool?

- C
- H
- A
- D
- S₂
- V
- A₂
- Sc

51 yo F with no other PMH

CHADS-VASc 1: <1%/yr
HAS-BLED 0: <1%/yr

Bleeding risk tool?

- H
- A
- S
- B
- L
- E
- D

68 yo M with HFpEF, HTN, DM, and CKD

CHADS-VASc 4: 5-7%/yr
HAS-BLED 3: ~6%/yr

[Calculate](#)
annual risk

Stroke Prevention

Risk Assessment

Therapies

Clinical Scenarios

Rate Control

Rhythm Control

Cases

Stroke Prevention: Therapies

When to anticoagulate?

CHADS ₂ -VA ₂ Sc	Stroke risk/y
0	0.2%
1	0.6%
2	2.2%
3	3.2%
4	4.8%
5	7.2%
6	9.7%
7	11%
8	11%
9	12%

1st line therapy?

DOACs > warfarin
(for most patients)

Paroxysmal vs. persistent?

AF burden (or prior cardioversion) does not change
anticoagulation decision

High bleeding risk?

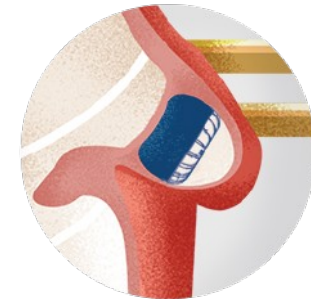
Address modifiable RFs.
Weigh risk/benefits of OAC.



OAC ↓ stroke 64%, ↓ mortality 26%
Significance of stroke often > major bleed

Non-medication options?

Percutaneous LAA occlusion
Surgical LAA ligation



Stroke Prevention

Risk Assessment

Therapies

Clinical Scenarios

Rate Control

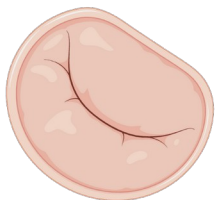
Rhythm Control

Cases

Anticoagulation: Clinical Scenarios

What if your patient has...

Valve disease



Moderate to severe MS or mechanical valve

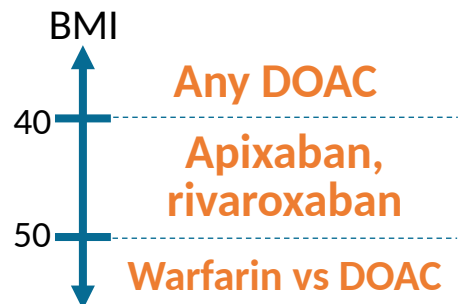
Warfarin

Frequent falls

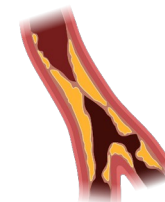


Continue OAC unless history of ICH

Obesity

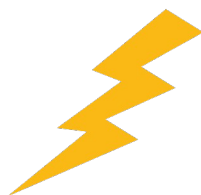


CAD



In general, do not need ASA + OAC long term for stable CAD

Cardioversion



AF > 48h or unknown

OAC x 3w or TEE DCCV

OAC x 4w regardless of risk

CKD



Apixaban or warfarin if ESRD or CrCl < 15

Stroke Prevention

Rate Control

Rhythm Control

Cases

Rate Control

What is the target HR?

Goal HR <110 for most patients

Stricter control (or rhythm control) if symptoms or declining LV function

Options for Rate Control?

Beta blockers

Medication

Metoprolol
Carvedilol
Bisoprolol

Dose

TDD 50-200 mg
TDD 6.25-50 mg
TDD 1.25-20 mg

Considerations

Preferred in stable HFrEF
Caution in severe
bronchospasm

CCBs

Diltiazem

TDD 120-480 mg

Avoid in HFrEF

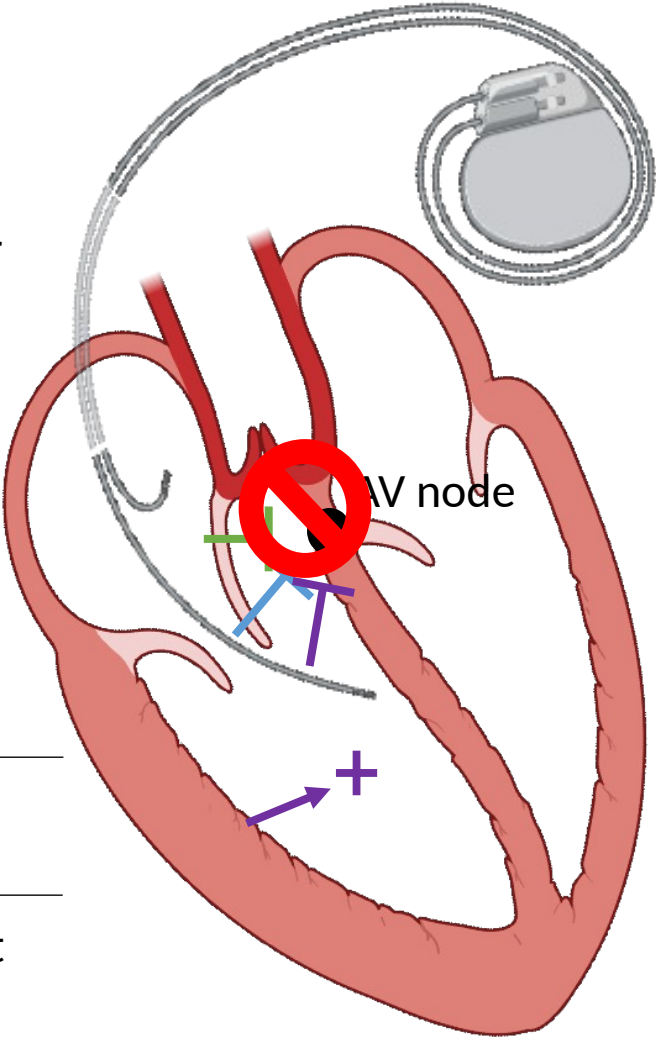
Digoxin

125-250 mcg daily

Second line, use as adjunct
Caution in CKD
Positive inotrope

Procedural option?
(Last line)

AV node ablation with biventricular pacemaker placement (CRT)



Rhythm Control

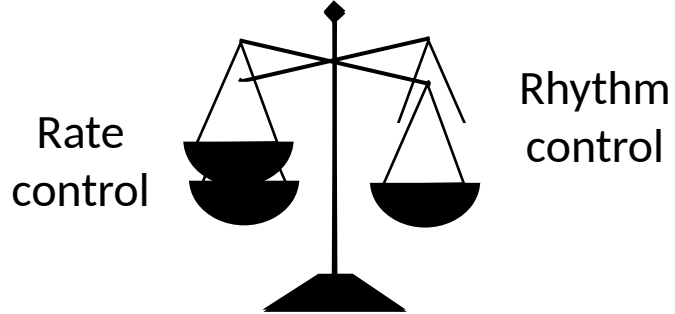
Stroke Prevention

Rate Control

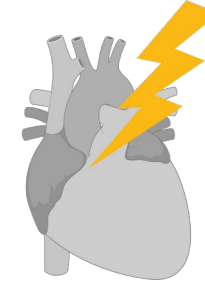
Rhythm Control

Cases

Historically...



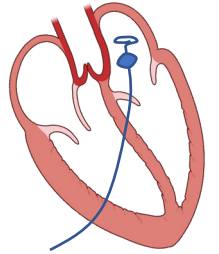
Therapies?



Cardioversion



Anti-arrhythmic medications



Ablation

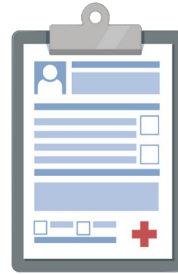
Factors that favor rhythm control?

Clinical Features

- Symptomatic
- Recent AF
- Failed rate control
- Tachycardia



Patient History



Risk reduction?



Smoking and ETOH cessation



Physical activity
Lose weight



Treat RFs
(DM, HTN, HLD, OSA)

Medications

Toxicity/Monitoring

Amiodarone

Pulmonary, thyroid, liver, drug interaction

Sotalol

QTc / TdP

Dofetilide

Renally dose

Flecainide

Only if no structural heart disease

Stroke Prevention

Rate Control

Rhythm Control

Cases

1

2-3

Case 1a

59 yo M with HLD, OA on chronic NSAIDs presents with dyspnea on exertion. On exam, he is found to have AF with a HR of 115 bpm. BMI is 45. TTE shows LVEF 58% without valvular disease.

Rate control?

Goal HR <110 bpm. A **beta-blocker** (e.g., metoprolol) or **calcium channel blocker** (e.g., diltiazem) can be used in this setting.

Rhythm control?

Besides new onset AF, he has no RFs that favor rhythm control (young age, few comorbidities). It is reasonable to trial rate control first.

Anticoagulation?

CHADS-VASc: 0. HAS-BLED 1 (NSAIDs).
Anticoagulation not indicated.

Case 1b

A few weeks later, he returns to clinic. HR at home has been in the 95-110 bpm. He has ongoing dyspnea.

Rate control?

Consider ↑ rate control to target ↓ HR, as this may help with symptoms.

Rhythm control?

Consider rhythm control given persistent symptoms.
Reduce risk by managing comorbidities (weight loss, statin for HLD, screen for OSA and substance use).

...He is scheduled for cardioversion.

Anticoagulation?

- Before cardioversion: 3 weeks of OAC or TEE with cardioversion
- After cardioversion: 4 weeks of OAC even though his CHADSVASC is 0
- Use apixaban or rivaroxaban due to BMI

Case 1c

Five years later, he is admitted for a GI bleed due to a gastric ulcer and intermittent AF.

Two years after this, he returns to clinic for a follow-up. He is now 66 yo. His vital show BP is 165/90 with normal HR. ECG confirms normal sinus rhythm. He denies any episodes of dyspnea.

Rate and rhythm control?

Currently in sinus rhythm. No rate control need unless he continues to have paroxysms of AF.

Start anticoagulation?

He has paroxysmal AF. There is risk of stroke. Annual stroke and bleeding risk is:

- CHADS-VASc 2 (age, HTN)
- HAS-BLED 4 (age, bleeding history, NSAID use, uncontrolled HTN)

Shared decision making. Consider anticoagulation with close monitoring and assessment of modifiable RFs: BP control, stop NSAIDs, follow up with GI.

Stroke Prevention

Rate Control

Rhythm Control

Cases

1

2-3

Case 2

Choose a strategy for stroke prevention in the following patients with AF.

75 yo M with HTN and moderate-severe aortic stenosis

CHADs-VASc 3. Anticoagulate with any **DOAC**. Only mitral valve or mechanical valve disease is "valvular" AF.

82 yo F with HTN, HLD, CKD stage IV. Baseline Cr 2.4.

CHADS-VASc 4. Anticoagulate with **apixaban 2.5 mg BID** (preferred for CrCl < 15). Reduced dose of apixaban (2.5 vs 5 mg BID) is given if patients have any 2 of the following: age ≥80, weight ≤60 kg, or Cr ≥1.5.

62 yo M with severe obesity with BMI 55, DM type II, PAD.

CHADS-VASc 2. Anticoagulation is recommended. Given BMI > 50, discuss risks and benefits of warfarin vs. apixaban or rivaroxaban. There is limited data but his BMI is not an absolute contraindication to DOAC. Consult with a pharmacist.

78 yo F with prior CVA with left sided weakness on aspirin, CAD, and frequent falls. History of NSAID related PUD with GI bleeding requiring a transfusion.

CHADS-VASc 4 (4.8% annual stroke risk). HASBLED 4 (8.9% risk of bleeding). However, anticoagulation should still be considered. Risk of functional debility from stroke outweighs risk of nonmajor bleeding. Any DOAC could be considered.

Case 3

What would you use for rate control of AF in the following agents?

75 yo M with HFrEF (EF 38%)

Evidence based **beta-blocker**: metoprolol succinate, carvedilol, or bisoprolol

55 yo F with severe asthma and normal LV function

Calcium channel blocker, **diltiazem** most commonly used. Avoid beta-blocker in patients with severe airflow obstruction.

62 yo M with mildly decompensated HFrEF, on metoprolol succinate 50 mg daily. HR >120. Renal function normal.

Cautious **up-titration of beta-blocker vs.** addition of **digoxin** as an adjunct given HFrEF.

62 yo M above with HFrEF. Started on digoxin and increased metoprolol without improvement in HR. SBP 90-100.

Cardiology referral for consideration of rhythm control or AV node ablation with pacemaker placement. Hypotension limits uptitration of metoprolol and lack of rate control with digoxin.

2024 Update — 2023 ACC/AHA/ACCP/HRS AF Guideline

Staging: AF now staged 1 (at-risk) → 2 (pre-AF) → 3 (AF: 3A paroxysmal, 3B persistent, 3C long-standing, 3D post-ablation) → 4 (permanent). Replaces duration-only labels.

Anticoagulate by absolute risk: $\geq 2\%$ /yr stroke risk → OAC (Class 1) — $\approx \text{CHA}_2\text{DS}_2\text{-VASc} \geq 2$ men / ≥ 3 women; score 1(m)/2(f) reasonable (2a). Use the score to estimate risk, not as a rigid cutoff.

DOAC over warfarin (Class 1): warfarin only for mechanical valve or moderate–severe mitral stenosis. The term “nonvalvular AF” is de-emphasized.

Catheter ablation upgraded: Class 1 first-line rhythm control in selected symptomatic paroxysmal AF (younger, few comorbidities) AND in AF with HFrEF.

Lifestyle is a pillar: weight loss (target BMI < 27), alcohol reduction, BP control, OSA screening improve rate- and rhythm-control success.

LAAO (e.g., WATCHMAN): upgraded to Class 2a when there is a long-term contraindication to anticoagulation.

Take-Home Points

1. Use CHA₂DS₂-VASc to estimate annual stroke risk; anticoagulate at $\geq 2\%$ /yr (≥ 2 men / ≥ 3 women) and in ALL patients peri-cardioversion regardless of score.
2. DOACs are first-line for most AF. Apixaban is usable in high BMI and ESRD/CrCl < 15 . Warfarin only for mechanical valve or moderate-severe mitral stenosis.
3. Rate control to HR < 110 bpm for most (beta-blocker or non-dihydropyridine CCB; digoxin as adjunct, esp. in HFrEF). Diltiazem is contraindicated in HFrEF.
4. Persistent symptoms most strongly favor rhythm control; the 2023 guideline makes catheter ablation Class 1 first-line in selected patients and in AF + HFrEF.

Attribution & Optimization Changelog

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Optimized June 2026: original drew on the 2019 ACC/AHA/HRS + 2020 ESC guidelines. Added the 2023 ACC/AHA/ACCP/HRS update (AF staging; absolute-risk anticoagulation; DOAC Class 1; catheter ablation Class 1 first-line; LAAO 2a; lifestyle pillar) on the preceding slide and in the companion package.

Deck edit: append-only — original interactive animations, graphics, and slide order preserved; no factual errors found in the original content.

Companion package: HTML lesson hub, 1-page clinical reference (PDF), 16 Anki cards, 5 board-style MCQs, and an EMR dotphrase. Educational use only; verify dosing and current guidance before teaching.